Lesson 2.4 — Factoring Quadratics

Big idea: Factoring is reverse multiplying. The trick is finding two numbers with the right sum and product — or recognizing a special pattern.

Quick review

GCF first: always check for a common factor before anything else.
Simple trinomial (a = 1): $x^2 + bx + c$. Find two numbers that multiply to c and add to b.
Difference of squares: $a^2 – b^2 = (a + b)(a – b)$.
Leading coefficient ≠ 1: use the AC method (or grouping). Multiply a·c, find pair that adds to b, split middle term, group.

Worked example

Worked example. Factor $2x^2 + 7x + 3$.

AC = 2·3 = 6. Need pair summing to 7: 6 and 1. Split: $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$.

Warm-ups

1. Factor $x^2 + 9x + 18$.
2. Factor $x^2 – 25$.
3. Factor $3x^2 – 12x$.

Practice

1. Factor $x^2 – 11x + 28$.
2. Factor $4x^2 – 49$.
3. Factor $6x^2 + 11x – 10$.

Challenge

1. Factor completely: $2x^3 – 18x$.
2. Factor $9x^2 – 12x + 4$. Then explain why both factors are the same.

Answer key — Lesson 2.4

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $(x + 3)(x + 6)$.
2. $(x - 5)(x + 5)$.
3. $3x(x - 4)$.

Practice

1. $(x - 4)(x - 7)$.
2. $(2x - 7)(2x + 7)$.
3. AC = –60; pair 15 and –4. $6x^2 + 15x - 4x - 10 = 3x(2x + 5) - 2(2x + 5) = (3x - 2)(2x + 5)$.

Challenge

1. GCF: $2x$. $2x(x^2 - 9) = 2x(x - 3)(x + 3)$.
2. $(3x - 2)(3x - 2) = (3x - 2)^2$. It's a perfect square trinomial: $a^2 - 2ab + b^2$ with $a = 3x$, $b = 2$.